AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

FACULTY OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

COURSE: INTRODUCTION TO DATA SCIENCE

FALL 2025-2026

Section: J

Group – M

Supervised By

Kamrun Naher koli

MID TERM PROJECT - 1

Name	Id
MD.MAKAIL ISLAM RAHIM	22-48946-3
SAIED AL MASUD	22-48943-3
HASIBUL ISLAM HASIB	22-48507-3

Dataset Details:

The dataset used in this project contains information about startups, including their operational status, total funding received in US dollars, and the number of funding rounds completed. The status column originally included operating, closed, acquired, and IPO, but acquired and IPO were grouped into operating for analysis. The dataset includes both numerical and categorical values and contains missing values, outliers, and inconsistencies, making it suitable for demonstrating data cleaning and predictive analysis.

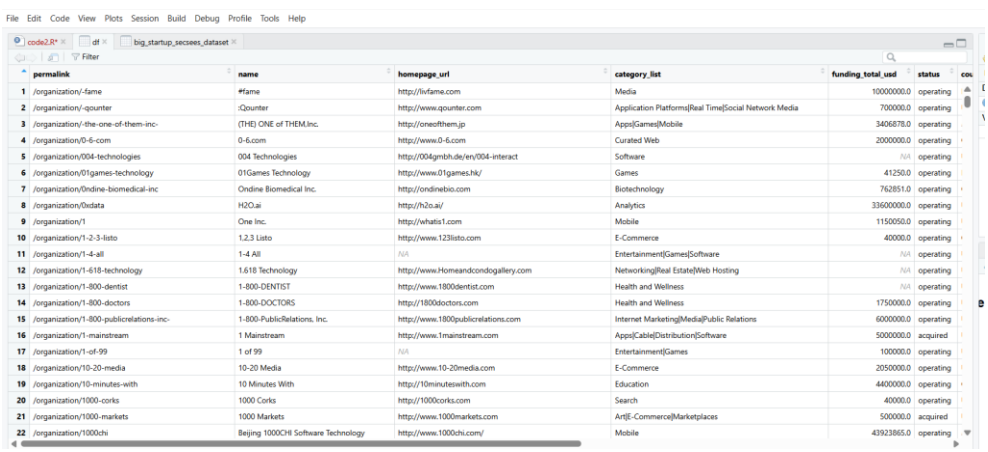
Loading the Library and Importing the Dataset :

R Code:

```
library(dplyr)

df <- read.csv("big_startup_secsees_dataset.csv", na.strings = c("", "-", "NA"))
View(df)
```

Output:



	permalink	name	homepage_url	category_list	funding_total_usd	status
1	/organization/-fame	#fame	http://livfame.com	Media	10000000.0	operating
2	/organization/-qounter	:Qounter	http://www.qounter.com	Application Platforms|Real Time|Social Network Media	700000.0	operating
3	/organization/-the-one-of-them-inc-	(THE) ONE of THEM,Inc.	http://oneofthem.jp	Apps|Games|Mobile	3406878.0	operating
4	/organization/0-6-com	0-6.com	http://www.0-6.com	Curated Web	2000000.0	operating
5	/organization/004-technologies	004 Technologies	http://004gmbh.de/en/004-interact	Software	NA	operating
6	/organization/01games-technology	01Games Technology	http://www.01games.hk/	Games	41250.0	operating
7	/organization/0ndine-biomedical-inc	Ondine Biomedical Inc.	http://ondinebio.com	Biotechnology	762851.0	operating
8	/organization/0xdata	H2O.ai	http://h2o.ai/	Analytics	33600000.0	operating
9	/organization/1	One Inc.	http://whatis1.com	Mobile	1150050.0	operating
10	/organization/1-2-3-listo	1,2,3 Listo	http://www.123listo.com	E-Commerce	40000.0	operating
11	/organization/1-4-all	1-4 All	NA	Entertainment|Games|Software	NA	operating
12	/organization/1-618-technology	1.618 Technology	http://www.Homeandcondogallery.com	Networking|Real Estate|Web Hosting	NA	operating
13	/organization/1-800-dentist	1-800-DENTIST	http://www.1800dentist.com	Health and Wellness	NA	operating
14	/organization/1-800-doctors	1-800-DOCTORS	http://1800doctors.com	Health and Wellness	1750000.0	operating
15	/organization/1-800-publicrelations-inc-	1-800-PublicRelations, Inc.	http://www.1800publicrelations.com	Internet Marketing|Media|Public Relations	6000000.0	operating
16	/organization/1-mainstream	1 Mainstream	http://www.1mainstream.com	Apps|Cable|Distribution|Software	5000000.0	acquired
17	/organization/1-of-99	1 of 99	NA	Entertainment|Games	100000.0	operating
18	/organization/10-20-media	10-20 Media	http://www.10-20media.com	E-Commerce	2050000.0	operating
19	/organization/10-minutes-with	10 Minutes With	http://10minuteswith.com	Education	4400000.0	operating
20	/organization/1000-corks	1000 Corks	http://1000corks.com	Search	40000.0	operating
21	/organization/1000-markets	1000 Markets	http://www.1000markets.com	Art|E-Commerce|Marketplaces	500000.0	acquired
22	/organization/1000chi	Beijing 1000CHI Software Technology	http://www.1000chi.com/	Mobile	43923865.0	operating

Description:

The dplyr library is loaded to enable efficient data manipulation, after which the dataset is imported using the read.csv() function and stored in the dataframe df. During import, blank entries, hyphens, and the string “NA” are treated as missing values to ensure accurate identification of incomplete data. The displayed preview of the dataset confirms that the file has been successfully loaded and shows the presence of key variables such as company name, category, funding amount, and operational status, along with some missing values that will require further cleaning.

Initial Data Preview:

R Code:

```
print(head(df))
```

Output:

```
Console   Terminal ×   Background Jobs ×
R 4.5.1 · E:/datascience/
> print(head(df))
                          permalink                    name                      homepage_url                                      category_list
1                 /organization/-fame                  #fame                http://livfame.com                                              Media
2              /organization/-qounter                :Qounter            http://www.qounter.com Application Platforms|Real Time|Social Network Media
3 /organization/-the-one-of-them-inc- (THE) ONE of THEM,Inc.                http://oneofthem.jp                                 Apps|Games|Mobile
4                /organization/0-6-com                  0-6.com                http://www.0-6.com                                       Curated Web
5      /organization/004-technologies       004 Technologies http://004gmbh.de/en/004-interact                                          Software
6    /organization/01games-technology     01Games Technology            http://www.01games.hk/                                             Games
  funding_total_usd    status country_code state_code                region          city funding_rounds founded_at first_funding_at last_funding_at
1          10000000 operating          IND         16                Mumbai        Mumbai              1       <NA>       2015-01-05      2015-01-05
2            700000 operating          USA         DE            DE - Other Delaware City              2 2014-09-04       2014-03-01      2014-10-14
3           3406878 operating         <NA>       <NA>                  <NA>          <NA>              1       <NA>       2014-01-30      2014-01-30
4           2000000 operating          CHN         22               Beijing       Beijing              1 2007-01-01       2008-03-19      2008-03-19
5                NA operating          USA         IL Springfield, Illinois     Champaign              1 2010-01-01       2014-07-24      2014-07-24
6             41250 operating          HKG       <NA>             Hong Kong     Hong Kong              1       <NA>       2014-07-01      2014-07-01
>
```

Description:

The print(head(df)) function displays the first six records of the dataset, providing a concise overview of its structure. The output reveals that each entry contains multiple attributes, including the startup's permalink, name, homepage URL, category classifications, funding amount, operating status, and additional geographic and temporal fields. The preview also indicates the presence of missing values in several columns, demonstrating the need for data cleaning and preprocessing.

Column Names and Missing Value :

R code:

```
colnames(df)
colSums(is.na(df))
```

Output:

```
> colnames(df)
 [1] "permalink"         "name"              "homepage_url"      "category_list"     "funding_total_usd" "status"            "country_code"      "state_code"
 [9] "region"            "city"              "funding_rounds"    "founded_at"        "first_funding_at"  "last_funding_at"
> colSums(is.na(df))
       permalink              name      homepage_url     category_list funding_total_usd            status      country_code        state_code            region
               0                 0              5058              3148             12785                 0              6958              8547              8030
            city    funding_rounds        founded_at  first_funding_at   last_funding_at
            8028                 0             15221                24                 0
>
```

Description:

The commands are used to inspect the structure of the dataset and evaluate the extent of missing data. The first output confirms the number of variables present in the dataset, while the

second reports the total number of missing values in each variable. The results indicate that several variables contain a substantial amount of missing data.

Selecting Variables and Checking Missing Data

R code:

```
df <- df %>% select(status, funding_total_usd, funding_rounds)
colnames(df)
colSums(is.na(df))
print(head(df))
```

Output:

```
> df <- df %>% select(status, funding_total_usd, funding_rounds)
> colnames(df)
[1] "status"            "funding_total_usd" "funding_rounds"
> colSums(is.na(df))
           status funding_total_usd    funding_rounds
                0             12785                 0
> print(head(df))
     status funding_total_usd funding_rounds
1 operating          10000000              1
2 operating            700000              2
3 operating           3406878              1
4 operating           2000000              1
5 operating                NA              1
6 operating             41250              1
>
```

Description:

In this step, only the variables required for the analysis are selected, allowing the dataset to be reduced to its most relevant components. After selecting these variables, their names are retrieved to confirm that the dataset has been correctly refined. A missing value check is then performed, revealing that one of the numerical variables contains a substantial number of missing entries. Finally, the first few rows of the updated dataset are displayed to verify that the selected variables have been extracted successfully and to provide an initial view of the remaining data.

Updated Status and Success Variable

R Code:

```

df$status[df$status == "acquired"] <- "operating"
df$status[df$status == "ipo"] <- "operating"
df$status <- as.factor(df$status)
df$success <- ifelse(df$status == "operating", 1, 0)
View(df)
```

Output:

	status	funding_total_usd	funding_rounds	success
1	operating	10000000.0	1	1
2	operating	700000.0	2	1
3	operating	3406878.0	1	1
4	operating	2000000.0	1	1
5	operating	NA	1	1
6	operating	41250.0	1	1
7	operating	762851.0	2	1
8	operating	33600000.0	4	1
9	operating	1150050.0	3	1
10	operating	40000.0	1	1
11	operating	NA	1	1
12	operating	NA	1	1
13	operating	NA	1	1
14	operating	1750000.0	1	1
15	operating	6000000.0	1	1
16	operating	5000000.0	1	1
17	operating	100000.0	1	1
18	operating	2050000.0	4	1
19	operating	4400000.0	2	1
20	operating	40000.0	1	1

Description:

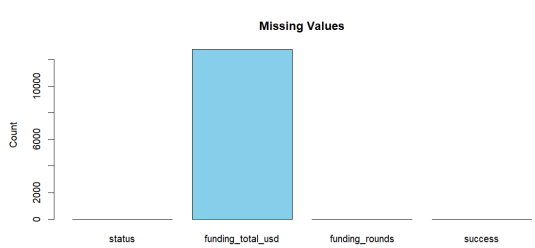
The status variable is standardized by merging alternative positive outcomes into a single category to ensure consistent classification. It is then converted into a factor, and a new binary variable, success, is created to distinguish operating startups from others. The updated output confirms that the recoding and binary assignment were applied correctly.

Missing Value Visualization

R Code:

```
missing_counts <- colSums(is.na(df))
barplot(missing_counts, main = "Missing Values", col = "skyblue", ylab = "Count")
```

Output:



Description:

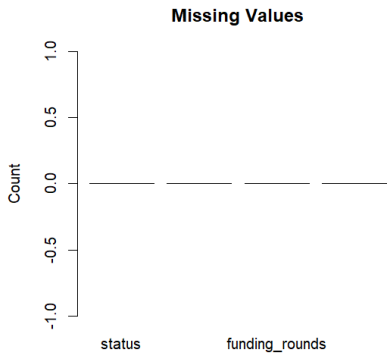
The barplot illustrates the number of missing values present in each selected variable. The visual clearly indicates that one of the numerical variables contains a substantial amount of missing data, while the remaining variables show no missing entries.

Missing Value Handling

R Code:

```
df$funding_total_usd[is.na(df$funding_total_usd)] <- mean(df$funding_total_usd, na.rm = TRUE)
df$funding_rounds[is.na(df$funding_rounds)] <- mean(df$funding_rounds, na.rm = TRUE)
colSums(is.na(df))
missing_counts <- colSums(is.na(df))
barplot(missing_counts, main = "Missing Values", col = "skyblue", ylab = "Count")
```

Output:



Description:

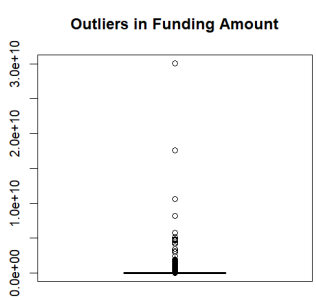
After applying mean imputation to replace missing numerical entries, the updated barplot shows that all variables now contain zero missing values. This confirms that the imputation process successfully addressed all incomplete observations. The dataset now represents a fully populated subset of the selected variables, allowing for more reliable statistical interpretation.

Outlier Detection

R Code:

```
boxplot(df$funding_total_usd, main = "Outliers in Funding Amount", col = "orange")
```

Output:



Description:

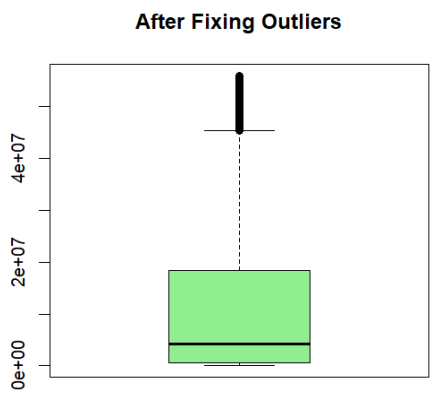
The boxplot illustrates the distribution of the funding amount and highlights the presence of several extreme values. These outliers appear far above the main concentration of observations, indicating unusually high funding amounts compared to the rest of the dataset. Identifying these extreme points is important because they can distort measures of central tendency and variability.

Outlier Handling

R Code:

```
upper <- quantile(df$funding_total_usd, 0.95)
lower <- quantile(df$funding_total_usd, 0.05)
df$funding_total_usd[df$funding_total_usd > upper] <- upper
df$funding_total_usd[df$funding_total_usd < lower] <- lower
boxplot(df$funding_total_usd, main = "After Fixing Outliers",col = "lightgreen")
```

Output:



Description:

The updated boxplot displays the distribution of the funding amount after applying percentile-based capping to limit extreme values. The previously extreme observations have been adjusted to fall within the defined upper and lower thresholds, resulting in a more balanced and stable distribution. This correction reduces the influence of unusually large or small values, improving the reliability of subsequent statistical analysis.

Success Group Comparison

R Code:

```r
tapply(df$funding_total_usd, df$success, summary)
tapply(df$funding_rounds, df$success, summary)
```

Output:

```
                    Max.   :56000000   Max.   :19.000   Max.
> tapply(df$funding_total_usd, df$success, summary)
$`0`
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
   37303   500000  2200000  8016443 10200000 56000000

$`1`
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
   37303   700000  3052000  8864597 11512945 56000000

> tapply(df$funding_rounds, df$success, summary)
$`0`
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1.0     1.0     1.0     1.9     2.0    14.0

$`1`
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   1.000   2.000   2.425   3.000  19.000
```

Description:

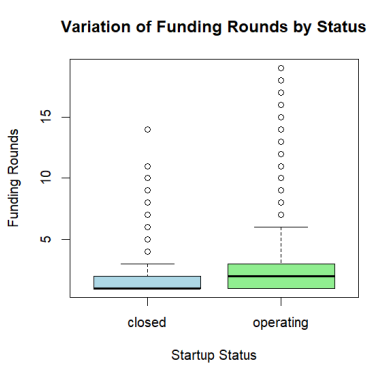
The grouped summary statistics compare funding amounts and funding rounds between successful and unsuccessful startups. The results show clear differences in central tendency and distribution across the two groups. Successful startups tend to exhibit higher median and mean funding levels, as well as slightly higher funding round activity. These statistics provide an initial indication that funding characteristics may be associated with the likelihood of success within the dataset.

Funding Rounds by Status

R Code:

```r
boxplot(df$funding_rounds ~ df$status,
        main = "Variation of Funding Rounds by Status",
        xlab = "Startup Status",
        ylab = "Funding Rounds",
        col = c("lightblue", "lightgreen"))
```

Output:



Description:

The boxplot compares the distribution of funding rounds between closed and operating startups. The visual shows that operating startups generally have higher median funding rounds and a wider spread of values, including more extreme observations. In contrast, closed startups display lower central values with a narrower distribution. This indicates noticeable variation in funding activity across the two status categories.